

Task 1

```
richie@richie-Latitude-5490:~/Azubi Africa/Group Activities$ docker_project_apacheHTTP
cd docker_project_apacheHTTP
docker_project_apacheHTTP: command not found
bash: cd: docker_project_apacheHTTP: No such file or directory
richie@richie-Latitude-5490:~/Azubi Africa/Group Activities$ mkdir docker_project_apacheHTTP
richie@richie-Latitude-5490:~/Azubi Africa/Group Activities$ cd docker_project_apacheHTTP
richie@richie-Latitude-5490:~/Azubi Africa/Group Activities/docker_project_apacheHTTP$ touch Dockerfile
richie@richie-Latitude-5490:~/Azubi Africa/Group Activities/docker_project_apacheHTTP$ nano Dockerfile
richie@richie-Latitude-5490:~/Azubi Africa/Group Activities/docker_project_apacheHTTP$ ls
Dockerfile
```

Task 2

```
GNU nano 7.2
# Use official Ubuntu base image
FROM ubuntu:22.04

# Prevent interactive prompts during installation
ENV DEBIAN_FRONTEND=noninteractive

# Update package list and install Python and necessary dependencies
RUN apt-get update && \
    apt-get install -y \
    python3 \
    python3-pip \
    python3-venv \
    && rm -rf /var/lib/apt/lists/*

# Create a symbolic link for python
RUN ln -s /usr/bin/python3 /usr/bin/python

# Create working directory
WORKDIR /app

# Create a basic Python script
RUN echo '#!/usr/bin/env python3\n\
print("Hello from Docker container!")\n\
print("Container is running successfully!")\n\
print("Group Name: Apache HTTP")\n\
print("Python version:")\n\
import sys\n\
print(sys.version)' > /app/hello.py

# Make the script executable
RUN chmod +x /app/hello.py

# Define the entry point
ENTRYPOINT ["python3", "/app/hello.py"]
```

Task 3

```
richie@richie-Latitude-5490: ~/Azubi Africa/Group Activities/docker_project_apacheHTTP$ docker build -t azdockerimageapachehttp .
[+] Building 118.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 894B
=> [internal] load metadata for docker.io/library/ubuntu:22.04
=> [internal] load .dockerignore
=> => transferring context: 28
=> [1/6] FROM docker.io/library/ubuntu:22.04@sha256:4f838adc7181d9039ac795a7d0aba05a9bd9ecd480d294483169c5def983b64d
=> => resolve docker.io/library/ubuntu:22.04@sha256:4f838adc7181d9039ac795a7d0aba05a9bd9ecd480d294483169c5def983b64d
=> => sha256:4f838adc7181d9039ac795a7d0aba05a9bd9ecd480d294483169c5def983b64d 6.69kB / 6.69kB
=> => sha256:ce941a2a18bbb922e434d6d6d2b31e571a5c3826eaf6ada0a41dcc905bd2d906 424B / 424B
=> => sha256:86f1a8d7b38e7a014c249cf2ca573c8ff7ce3cca128c5c06dcee758813726f90 2.05kB / 2.05kB
=> => sha256:40d16f30db405106ef8074779bdf41f012465c2a785bbeaa2eab9f2081099b47 29.74MB / 29.74MB
=> => extracting sha256:40d16f30db405106ef8074779bdf41f012465c2a785bbeaa2eab9f2081099b47
=> [2/6] RUN apt-get update && apt-get install -y python3 python3-pip python3-venv && rm -rf /var/lib/apt/lists/*
=> [3/6] RUN ln -s /usr/bin/python3 /usr/bin/python
=> [4/6] WORKDIR /app
=> [5/6] RUN echo '#!/usr/bin/env python3\nprint("Hello from Docker container!")\nprint("Container is running successfully!")\nprint("Group Name: Apache")' > /app/hello.py
=> [6/6] RUN chmod +x /app/hello.py
=> exporting to image
=> => exporting layers
=> => writing image sha256:ce718cdddc45ae49580257239fa8dc78002120692773c5056f960cab2513e3ae
=> => naming to docker.io/library/azdockerimageapachehttp
richie@richie-Latitude-5490: ~/Azubi Africa/Group Activities/docker_project_apacheHTTP$
```

Task 4

```
Setting up python3-dev (3.10.6-1-22.04.1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
Processing triggers for ca-certificates (20240203-22.04.1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
root@e5db55d287f3:/# exit
exit
richie@richie-Latitude-5490: ~/Azubi Africa/Group Activities/docker_project_apacheHTTP$ docker commit temp_container azdockerimagegroupname_committed
^Crichie@richie-Latitude-5490: ~/Azubi Africa/Group Activities/docker_project_apacheHTTP$ docker commit temp_container azdockerimageapachehttp_committed
sha256:ddf5099cddb764c18c0a6862a70b834737e7e113a3df4c7ecd7820e2c5b2a252
richie@richie-Latitude-5490: ~/Azubi Africa/Group Activities/docker_project_apacheHTTP$ docker images | grep azdockerimageapachehttp_committed
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
azdockerimageapachehttp_committed:latest ddf5099cddb7 504MB 0B
richie@richie-Latitude-5490: ~/Azubi Africa/Group Activities/docker_project_apacheHTTP$ docker run -it --rm azdockerimageapachehttp_committed python3 -c "print('Python installed successfully!')"
bash: !': event not found
richie@richie-Latitude-5490: ~/Azubi Africa/Group Activities/docker_project_apacheHTTP$ docker run -it --rm azdockerimageapachehttp_committed python3 -c "print('Python installed successfully!')"
Python installed successfully!
richie@richie-Latitude-5490: ~/Azubi Africa/Group Activities/docker_project_apacheHTTP$
```